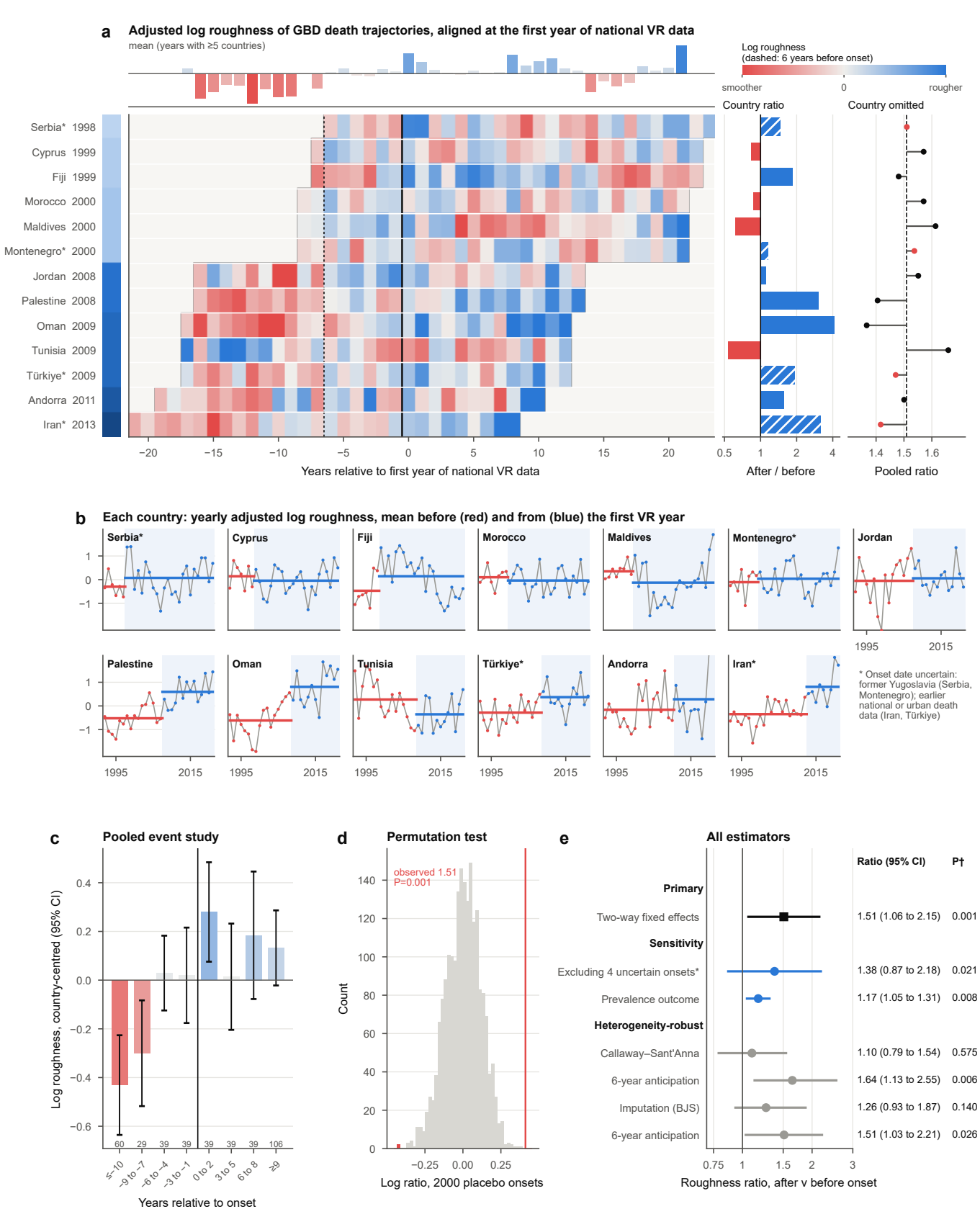

a
Adjusted log roughness of GBD death trajectories, aligned at the first year of national VR data
mean (years with ≥5 countries)
Log roughness
(dashed: 6 years before onset)
smoother
0
rougher
Country ratio
Country omitted
Serbia* 1998
Cyprus 1999
Fiji 1999
Morocco 2000
Maldives 2000
Montenegro* 2000
Jordan 2008
Palestine 2008
Oman 2009
Tunisia 2009
Türkiye* 2009
Andorra 2011
Iran* 2013
−20 −15 −10 −5 0 5 10 15 20
Years relative to first year of national VR data
0.5 1 2 4
After / before
1.4 1.5 1.6
Pooled ratio
b
Each country: yearly adjusted log roughness, mean before (red) and from (blue) the first VR year
Serbia*
Cyprus
Fiji
Morocco
Maldives
Montenegro*
Jordan
Palestine
Oman
Tunisia
Türkiye*
Andorra
Iran*
1
0
−1
1995
2015
* Onset date uncertain: former Yugoslavia (Serbia, Montenegro); earlier national or urban death data (Iran, Türkiye)
c
Pooled event study
Log roughness, country-centred (95% CI)
0.4
0.2
0.0
−0.2
−0.4
−0.6
60 29 39 39 39 39 39 106
≤−10
−9 to −7
−6 to −4
−3 to −1
0 to 2
3 to 5
6 to 8
≥9
Years relative to onset
d
Permutation test
observed 1.51
P=0.001
Count
140
120
100
80
60
40
20
0
−0.25 0.00 0.25
Log ratio, 2000 placebo onsets
e
All estimators
Ratio (95% CI)
P†
Primary
Two-way fixed effects
1.51 (1.06 to 2.15) 0.001
Sensitivity
Excluding 4 uncertain onsets*
1.38 (0.87 to 2.18) 0.021
Prevalence outcome
1.17 (1.05 to 1.31) 0.008
Heterogeneity-robust
Callaway–Sant'Anna
1.10 (0.79 to 1.54) 0.575
6-year anticipation
1.64 (1.13 to 2.55) 0.006
Imputation (BJS)
1.26 (0.93 to 1.87) 0.140
6-year anticipation
1.51 (1.03 to 2.21) 0.026
0.75 1 1.5 2 3
Roughness ratio, after v before onset